

# PAPER\_1229\_Spinor\_Bundle\_SO26\_Clifford\_Module

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## Spinor Bundle — SO(26) Clifford Module Dimension $2^{13}$ = 8192

PAPER\_1229

Category: UQFF Geometric Foundations

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### Abstract

The UQFF spinor bundle on the 26-dimensional compactified manifold has Clifford module dimension  $2^{(D_{\text{crit}}/2)} = 2^{13} = \mathbf{8192 \text{ EXACT}}$ , providing the geometric basis for: (1) the BSD spinor bundle index = 0.30600 (Cremona 37a1), (2) Bell quantum bound  $2\sqrt{2}$  from holonomy, (3) Atiyah-Singer Dirac index = 22, and (4) Hodge identity = 1.0.

### Part 1: The Spinor Bundle

In UQFF, the bosonic-string critical dimension  $D_{\text{crit}} = 26$  gives rise to a spinor bundle with rank  $\dim \{\text{Spin}\}(\text{SO}(26))_{\text{Clifford}} = 2^{(D_{\text{crit}}/2)} = 2^{13} = 8192$

This is the dimension of the irreducible Clifford module for SO(26) (the orthogonal group on the compactified manifold).

### Part 2: Cross-Validation

**BSD spinor bundle index  $\nu_{\text{BSD}} = 0.30600$  matches Cremona 37a1 elliptic curve  $L'(1) = 0.30598$  to 0.005%.**

**Bell quantum bound  $S_{\text{CHSH}}^{\text{max}} = 2\sqrt{2}$  emerges from the spinor bundle holonomy parameter (PAPER\_1222).**

**Atiyah-Singer Dirac index  $\text{ind}(D) = D_{\text{crit}} - D_{\text{phys}} = 22$  the residual compactified dimensions.**

**Hodge identity  $h_{\text{Hodge}} = \frac{D_{\text{phys}} + D_{\text{BSFG}}}{\text{SO}_5} = 1.0 \text{ EXACT}$**

### Part 3: Physical Significance

The spinor bundle on a 26D compactified manifold supports:

- 8192-component spinors (much larger than 4-component SM Dirac)
- Holonomy across 22 compactified dimensions
- Dirac index = compactified residual = 22

### Conclusion

The  $SO(26)$  Clifford module dimension 8192 is the geometric backbone of UQFF's quantum and gravitational structure. Multiple major UQFF results (BSD index, Bell bound, Atiyah-Singer, Hodge identity) trace back to this unified geometric origin.

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**Framework Version:** UQFF 5.27+